Infosys_HotelRevAI – AI Driven Revenue Analysis For Hotels

Progress Report – Milestone 3

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1. Abstract

The hospitality industry generates large volumes of data through hotel bookings, customer interactions, pricing strategies, and operational activities. However, raw data alone does not provide meaningful insights unless it is properly analyzed and visualized. Business intelligence tools help organizations convert raw data into actionable insights that support effective decision making. This project focuses on analyzing hotel booking data using Power BI to understand revenue trends, customer behavior, and operational performance in the hotel industry.

The dataset used in this project contains hotel booking information from the years **2018 to 2020**. The data includes various attributes such as hotel type, customer details, booking channels, market segments, stay duration, and pricing information. Before performing the analysis, the dataset was cleaned and transformed to ensure accuracy and consistency. Various calculated metrics such as total revenue, total bookings, average daily rate, and average length of stay were created to analyze hotel performance.

The project consists of four interactive dashboards: **Executive Overview, Booking Behavior Analysis, Revenue & Pricing Analysis, and Operational Insights**. Each dashboard focuses on a different aspect of hotel operations and provides valuable insights through visualizations. These dashboards help hotel management understand booking trends, evaluate pricing strategies, analyze customer behavior, and improve operational efficiency.

The results of the analysis reveal several key insights. Online travel agencies contribute significantly to hotel bookings, most guests are new customers rather than repeat visitors, and seasonal trends have a strong impact on hotel revenue. The dashboards created in Power BI enable users to explore the data interactively and gain deeper insights into hotel performance. Overall, the project demonstrates how data analytics and visualization can support data-driven decision making in the hospitality industry.

2. Introduction

The hospitality industry plays a significant role in the global economy, with hotels serving millions of guests every year. Hotels generate vast amounts of data through reservation systems, customer records, and financial transactions. This data includes valuable information about customer preferences, booking patterns, pricing strategies, and operational performance. However, analyzing this data manually can be time-consuming and inefficient.

Business intelligence tools such as **Power BI** allow organizations to analyze large datasets and transform them into meaningful visual insights. Through interactive dashboards and reports, decision makers can easily monitor key performance indicators and identify trends that affect business performance. Data visualization helps simplify complex datasets and makes it easier for management to understand business performance.

This project focuses on the development of a **Hotel Revenue Analysis Dashboard** using Power BI. The aim is to analyze hotel booking data and present the results through interactive dashboards. These dashboards provide insights into revenue trends, customer booking behavior, pricing strategies, and operational performance.

By analyzing historical hotel booking data, this project helps identify patterns that can improve hotel management strategies. The use of data analytics allows hotel managers to make informed decisions related to pricing, marketing, and resource allocation. Ultimately, the project demonstrates the importance of using data-driven approaches to improve business performance in the hospitality sector.

3. Problem Statement

Hotels collect a large amount of booking data through reservation systems and customer interactions. This data includes important information such as customer demographics, booking channels, room pricing, and stay duration. However, many hotels struggle to convert this data into useful insights that support strategic decision making.

Without proper analysis, it becomes difficult for hotel management to track revenue performance, identify seasonal trends, and understand customer booking behavior. Lack of data-driven insights can result in inefficient pricing strategies, poor marketing decisions, and reduced customer satisfaction.

Therefore, there is a need for a data analytics solution that can transform raw hotel booking data into meaningful insights. By developing interactive dashboards using Power BI, hotel management can monitor business performance, analyze customer behavior, and make better decisions that improve revenue and operational efficiency.

4. Project Objectives

The primary objective of this project is to analyze hotel booking data and create interactive dashboards using Power BI. These dashboards aim to provide meaningful insights into hotel revenue performance and customer booking patterns.

Another objective of the project is to analyze customer behavior by studying booking channels, lead time distribution, customer types, and repeat guest patterns. Understanding these factors helps hotels improve their marketing strategies and increase customer retention.

The project also aims to evaluate hotel pricing strategies using metrics such as Average Daily Rate (ADR) and revenue trends. By analyzing how pricing affects booking patterns, hotel management can optimize pricing strategies to maximize revenue.

Additionally, the project focuses on analyzing operational aspects such as stay duration distribution, seasonal trends, and geographical distribution of guests. These insights help hotels improve resource planning and operational efficiency.

5. Dataset Description

The dataset used in this project contains hotel booking records from **2018 to 2020**. It includes detailed information related to hotel reservations, customer details, booking channels, and pricing information. This dataset provides a comprehensive view of hotel operations and customer booking behavior.

Key attributes in the dataset include **hotel type**, which distinguishes between City Hotels and Resort Hotels. The dataset also includes **market segment information**, which identifies the booking source such as Online Travel Agencies, Offline Travel Agencies, Direct bookings, Corporate bookings, and Group reservations. These attributes help analyze how customers book hotels and which channels generate the most revenue.

In addition to booking details, the dataset includes important revenue-related metrics such as **Average Daily Rate (ADR)** and the number of **weekday and weekend nights** spent by guests. These metrics help analyze pricing strategies and stay duration patterns.

The dataset also includes customer-related information such as **customer type, country of origin, and repeated guest status**. By analyzing these attributes, the dataset enables the study of booking trends, revenue generation patterns, and customer behavior in the hospitality industry.

6. Data Preprocessing and Transformation

Before building the dashboards, the dataset required several preprocessing steps to ensure data quality and accuracy. Raw datasets often contain missing values, duplicate records, and inconsistent data formats that can affect analysis results.

The first step in data preprocessing involved **data cleaning**, which included removing duplicate records and correcting inconsistent values. Missing values in certain attributes were handled carefully to maintain data reliability.

After cleaning the dataset, **data transformation** was performed to create calculated columns and measures required for analysis. Important metrics such as total revenue, total bookings, average daily rate, and average length of stay were calculated using Power BI.

These preprocessing steps ensured that the dataset was properly structured and ready for analysis. The cleaned dataset was then used to build interactive dashboards that provide meaningful insights into hotel performance.

7. Tools and Technologies Used

Several tools and technologies were used in this project to analyze and visualize hotel booking data. **Microsoft Excel** was used for initial data handling, data cleaning, and basic preprocessing tasks.

The main tool used in this project is **Microsoft Power BI**, which is a powerful business intelligence and data visualization platform. Power BI enables users to create interactive dashboards, perform data analysis, and generate insights through visual representations such as charts, graphs, and maps.

Business intelligence techniques were also applied to transform raw booking data into meaningful insights. These techniques help identify trends, patterns, and relationships within the dataset

8. Dashboard Architecture

To analyze the dataset effectively, the project includes four dashboards that focus on different aspects of hotel performance. Each dashboard provides unique insights that help management understand business operations more effectively.The dashboards included in this project are **Executive Overview Dashboard, Booking Behavior Analysis Dashboard, Revenue & Pricing Analysis Dashboard, and Operational Insights Dashboard**. Together, these dashboards provide a comprehensive analysis of hotel performance.Each dashboard contains multiple visualizations such as bar charts, line charts, pie charts, and maps. Interactive filters and slicers allow users to explore the data dynamically and analyze specific aspects of hotel performance.

9. Executive Overview Dashboard

The Executive Overview Dashboard provides a high-level summary of hotel performance. It displays key performance indicators such as **total revenue, total bookings, average daily rate, and average length of stay**.

The dashboard includes visualizations that show revenue trends across different years and booking trends over time. These visualizations help management understand overall business performance and identify growth patterns.

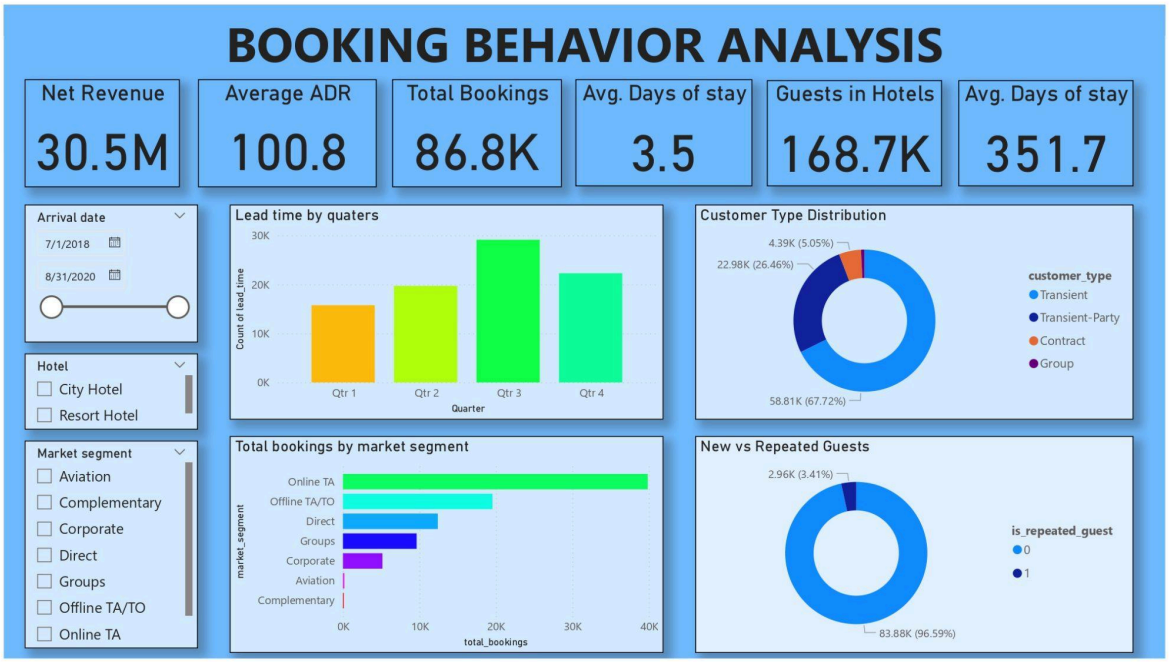
Another important feature of this dashboard is the comparison of revenue generated by **City Hotels and Resort Hotels**. This comparison helps management understand which hotel type contributes more to overall revenue.

10. Booking Behavior Analysis Dashboard

The Booking Behavior Analysis Dashboard focuses on understanding customer reservation patterns. It analyzes booking behavior using factors such as lead time distribution, customer type distribution, booking channels, and repeat guest patterns.

The analysis shows that the majority of hotel guests are new customers, while only a small percentage are repeat visitors. The dashboard also reveals that most bookings come from **transient customers**, who typically prefer short stays.

Additionally, the dashboard highlights booking distribution across different market segments such as online travel agencies, offline travel agencies, corporate bookings, and group bookings. These insights help hotels identify the most effective booking channels.



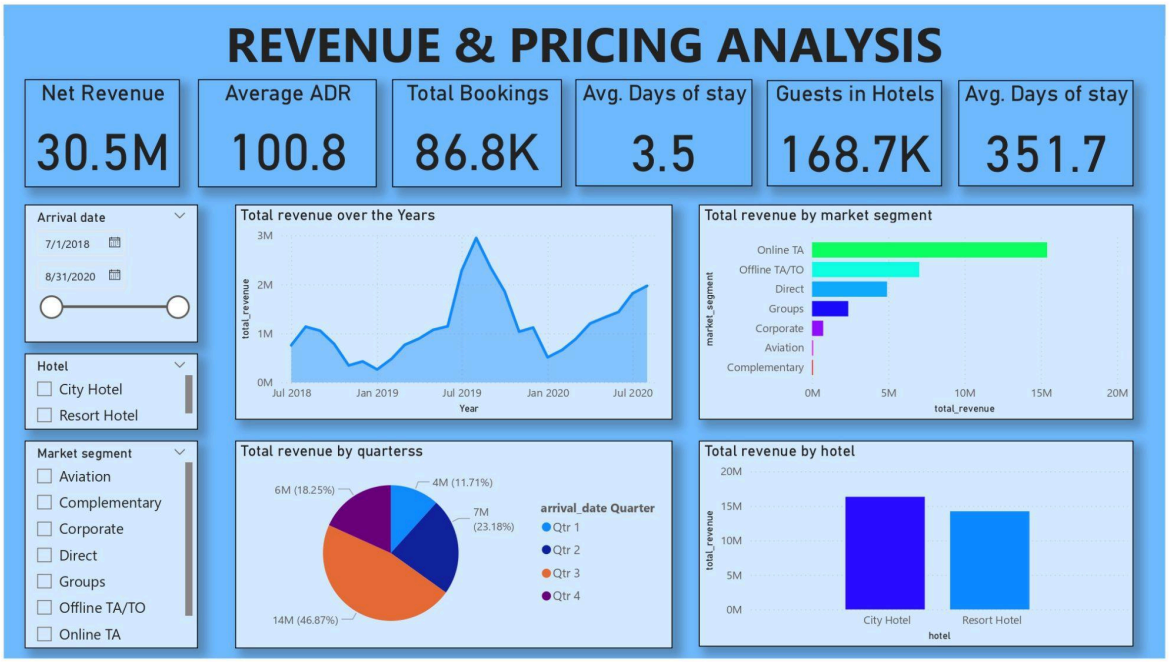
11. Revenue & Pricing Analysis Dashboard

This dashboard focuses on analyzing **hotel revenue performance and pricing strategies** over the given time period. It helps hotel management understand how revenue changes across different time intervals and how pricing decisions influence overall financial performance. The dashboard includes several visualizations that present revenue trends across **different quarters and years**, allowing analysts to observe seasonal variations and identify peak revenue periods. By studying these patterns, hotel managers can better understand when demand is highest and plan their pricing and promotional strategies accordingly.

In addition to time-based analysis, the dashboard also compares **revenue generated by different hotel types**, such as City Hotels and Resort Hotels. This comparison helps identify which type of hotel contributes more significantly to total revenue. The dashboard further analyzes **revenue distribution across various market segments**, including Online Travel Agencies, Offline Travel Agencies, Direct bookings, Corporate bookings, and Group reservations. This analysis allows hotels to determine which booking channels are the most profitable and which segments require more marketing focus.

Another important aspect of this dashboard is the evaluation of **pricing metrics such as the Average Daily Rate (ADR)**. ADR represents the average amount paid for rooms sold on a particular day and is a key indicator of pricing performance. By analyzing ADR along with revenue trends, hotels can assess whether their pricing strategies are effective in maximizing revenue. The insights gained from this dashboard help hotel management make informed

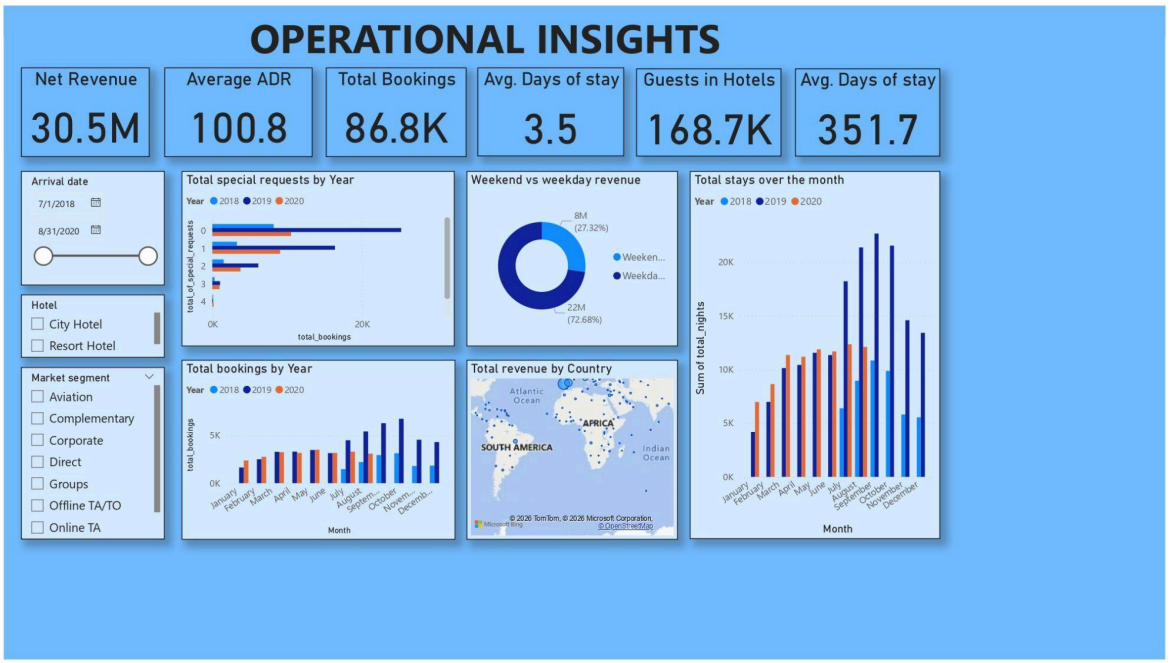
decisions regarding pricing adjustments, marketing strategies, and revenue optimization to improve overall profitability.



12. Operational Insights Dashboard

- The **Operational Insights Dashboard** focuses on analyzing the operational performance of the hotel and provides important information about how hotel services and resources are utilized. This dashboard helps hotel management understand various operational factors that affect day-to-day hotel activities. By studying these insights, hotels can improve their operational efficiency and provide better services to guests.
- One of the key analyses in this dashboard is the **stay duration distribution**, which shows how long guests typically stay at the hotel. It includes the number of **weekday nights and weekend nights** spent by guests. Understanding the average stay duration helps hotels manage room availability, housekeeping schedules, and other operational resources more effectively. The dashboard also analyzes **weekday versus weekend revenue**, which helps identify which time period generates higher revenue and when the hotel experiences higher occupancy.
- Another important feature of this dashboard is the analysis of **special guest requests**, such as additional services or preferences requested by guests during their stay. By studying these requests, hotels can better understand customer expectations and improve service quality. The dashboard also includes a **map visualization** that displays the geographical distribution of hotel guests based on their countries of origin. This helps hotels identify key regions that generate the most bookings and focus their marketing strategies accordingly.

Overall, the operational insights provided by this dashboard help hotel management make better decisions related to **resource planning, staffing management, and service improvement**, which ultimately enhances the overall guest experience and operational performance of the hote



13. Key Insights and Findings

- The analysis revealed several important insights about hotel performance and customer behavior. One major finding is that **online travel agencies contribute a significant portion of bookings and revenue**.
- Another key insight is that the majority of guests are **new customers**, indicating an opportunity for hotels to improve customer retention strategies.
- Seasonal trends also play an important role in hotel revenue generation. Certain periods experience higher booking demand due to travel seasons and holidays.

14. Conclusion

This project demonstrates how data analytics and visualization tools can transform hotel booking data into valuable insights. By using Power BI dashboards, hotel management can analyze revenue trends, understand customer behavior, and improve operational efficiency.The interactive dashboards developed in this project provide a clear overview of hotel performance and support data-driven decision making. These insights can help hotels optimize pricing strategies, improve marketing efforts, and enhance overall business performance.

15. Future Scope

- The project can be further improved by integrating **real-time hotel booking data**, which would allow management to monitor business performance continuously.
- Another possible improvement is the use of **machine learning models** to predict future booking trends and revenue patterns. Predictive analytics can help hotels prepare for peak seasons and optimize pricing strategies.
- Additionally, the dashboard can be enhanced by incorporating **customer feedback analysis and sentiment analysis** from online reviews. These insights can help hotels improve service quality and customer satisfaction.